# THANGARAJ T

*AI/ML Engineer | Machine Learning | Computer Vision*

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LinkedIn: Thangaraj T | GitHub: My AI-ML Tech Journey | Portfolio: URL pending deployment

## PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience in Python, Machine Learning, Deep Learning, Computer Vision, OCR/NLP and RAG-based Generative AI. Currently building ClearCustoms AI, an intelligent customs document processing solution, at Femtosoft Technologies; previously developed a face recognition attendance system at OneYes Infotech. Strengthening full-stack skills (HTML, CSS, JavaScript, React JS) to ship AI backends with complete frontends. M.E. (AI & ML) student, available immediately.

## TECHNICAL SKILLS

**Programming:** Python, SQL
**Machine Learning:** Scikit-learn, Regression, Classification, Clustering, Model Evaluation, Feature Engineering
**Deep Learning / GenAI:** TensorFlow/Keras, CNN, Transfer Learning (VGG16), LLMs, RAG, Gemini, Langchain.
**Document AI / NLP:** OCR, NLP, Document AI, Information Extraction & Validation.
**Computer Vision:** OpenCV, MTCNN, Face Detection & Recognition, Image Processing
**Backend & Frontend:** FastAPI, REST APIs, Streamlit, HTML, CSS, JavaScript, React JS (learning)
**Data & Tools:** NumPy, Pandas, Matplotlib, Seaborn, Pillow, python-docx, Git, GitHub, Render, Jupyter, VS Code | SQLite, SQLAlchemy

## EXPERIENCE

**AI/ML Intern | Femtosoft Technologies, Pallavaram, Chennai** *Sep 2026 – Present*

- **ClearCustoms AI:** building an intelligent customs document processing solution using OCR, NLP, LLMs and Document AI with Python, FastAPI and REST APIs.
- Extracting and validating information from customs documents and mapping the extracted data to the relevant ICEGATE/customs portal fields.
- Developing a complete frontend (Streamlit) integrated with the AI backend.

**AI/ML Developer Intern | OneYes Infotech Solutions, Chennai** *Aug 2026 – Sep 2026*

- Built a **Face Recognition Attendance System** using MTCNN for face detection, OpenCV-based preprocessing and face-recognition/attendance-recording workflows.
- Developed single-face and multi-face detection workflows, plus certificate and bulk-document generation utilities using Pillow, Streamlit, pandas and python-docx.

**AI/ML Intern | DataMites** *Jan 2026 – Jul 2026*

- Built an AI PDF Chatbot using RAG so users can query document content in natural language, and the ThangarajAI Chatbot, an AI/ML interview assistant.
- Developed Urban Heat Island Analysis & Prediction, modelling land surface temperature and urban density to predict heat island effects across local regions.
- Delivered end-to-end ML projects covering preprocessing, model development, evaluation, deployment and application integration.

## PROJECTS

- **Aqua Genesis – Agentic AI Flood Resilience Engine** (JCT SDG Hackathon 2026): four-agent swarm built with LlamaIndex Workflows, Gemini, SWMM and Streamlit.
- **AI Helpdesk Ticket Classifier:** TF-IDF + Logistic Regression, ~94% accuracy. **Credit Default Prediction:** SMOTE, StandardScaler, leak-free train/test split.

## EDUCATION

**M.E. – CSE (AI&ML), RVS College of Engineering and Technology.**

**B.E. – Electronics & Communication Engineering, Sri Sairam College of Engineering, Chennai**.

## ADDITIONAL INFORMATION

**Languages:** English, Tamil **Availability:** Immediate .